

Figure 8: Viewing data

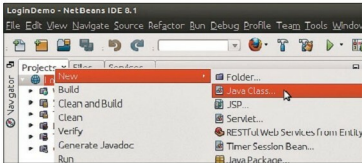


Figure 9: Creating a new resource

then create the password, first name and last name in the same way but do not select any constraint. At the end, close the *Create table* dialogue box by clicking on *OK*. We can see the table and column by expanding the *Table* option.

Right click on the login table and select *Execute command* to insert some dummy data. Once inserted, again right click on the login table and select *View data*, which will display the inserted data as shown in Figure 8.

The initial part of creating a database is completed and now we will start with application development.

Developing an application

Clicking on the *Project* tab will show our *LoginDemo* project. To create files in our Web application, select

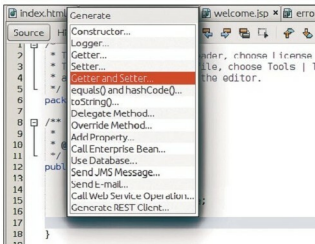


Figure 10: Inserting code

LoginDemo in *Projects*, and then select *New* which will display the options for various types of files/folders that can be created, as shown in Figure 9.

We can see the options for creating *Java Class*, *Servlet* and *JSP*, which are required for our MVC application.

Let's start with creating Beans. To do so, select *Java Class* which will ask for the parameters for *Class name*; enter that as *UserBean* and specify the package name.

Add three private variables, i.e., email, firstname, lastname.

Press *Enter* after lastname, press the right click button which will open the context menu, and select the *Insert code* option. It will display the option to generate *Constructor*, *Getter* and *Setter* methods as well as many other options, as shown in Figure 10.

Select *Getter* and *Setter*, select all fields and generate the *Getter* and *Setter* methods. Follow the same steps and click on *Constructor*. Create two *Constructors* — one without any argument and the other with three arguments.

The next step is to create *Servlet*, which will act as the controller. Again, right click on *Project name*, select *New Servlet*, which will ask for the servlet's name and package.

Click on *Next* and you will be asked to configure servlet deployment.

Select *Add information to deployment description*. In the URL pattern, change it to lower case. This will create the *web.xml* for mapping the servlet with the URL. We can leave it blank and annotation will be used for the same.

In *Servlet*, we will add the code for the database connection, fetching data, creating Beans, sharing the Beans and forwarding requests. You can download the servlet code from http://opensourceforu.com/article_source_code/may16/NetBeans.zip. In the *processrequest* method:

- Read the parameters, i.e., email and password, which come with *Request*.
- Connect with the database providing the necessary credentials that we created.
- Query the database for email and password by running a select query.

If query returns the data:

- Create the *UserBean* object with values from the database using the *Constructor*.
- Store the object in *Request*.
- Set the URL to *welcome.jsp*.

If the query does not return any data:

- Create the error message.
- Store the message in *Request*.
- Set the URL to *error.jsp*

In case of an exception:

- Create the error message.
- Store the message in *Request*.